

TOMPO, Russian Federation

WMO: 246710

Lat: 63.950N

Lon: 135.867E

Elev: 1318

StdP: 14.01

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-61.8	-59.2	-63.9	0.1	-60.8	-61.7	0.1	-58.9	19.0	-40.2	17.8	-45.0	4.5	350	0.767

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.1	83.4	63.1	79.4	61.3	75.6	59.7	65.5	78.4	63.3	75.3	61.3	72.4	5.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	84.2	69.3	58.9	77.8	67.0	56.7	71.9	65.7	31.1	78.7	29.5	75.2	28.0	73.1	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.0	10.0	DB	-63.8	89.3	2.7	4.1	-65.7	92.3	-67.3	94.7	-68.8	97.0	-70.8	100.0	(4)
(5)			WB	-63.6	68.8	2.8	2.4	-65.6	70.5	-67.2	71.9	-68.8	73.2	-70.8	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-43.7	-36.1	-13.4	15.2	40.7	56.1	61.2	54.5	38.3	9.7	-26.0	-42.7		
	DBStd	40.27	10.69	10.56	13.15	11.77	7.76	6.67	6.38	6.44	7.41	14.72	14.86	10.98		
	HDD50	15435	2903	2410	1965	1043	299	21	5	28	356	1249	2281	2875		
	HDD65	20213	3368	2830	2430	1493	752	274	149	331	801	1714	2731	3340		
	CDD50	740	0	0	0	0	11	204	353	167	5	0	0	0		
	CDD65	45	0	0	0	0	0	7	33	5	0	0	0	0		
	CDH74	1137	0	0	0	0	5	290	704	138	0	0	0	0		
	CDH80	285	0	0	0	0	0	63	200	22	0	0	0	0		
	WSAvg	4.2	4.7	4.7	4.5	4.5	4.3	4.0	3.7	3.3	3.8	3.8	4.6	4.4		
	PrecAvg	10.7	0.2	0.2	0.2	0.4	1.0	1.8	2.3	2.1	1.2	0.7	0.5	0.3		
	PrecMax	16.2	0.7	0.5	1.2	1.9	5.6	3.2	5.0	4.4	3.1	1.8	1.2	0.7		
(15) Precipitation	PrecMin	7.4	0.0	0.0	0.0	0.0	0.0	0.6	0.1	0.5	0.0	0.1	0.2	0.0		
	PrecStd	2.0	0.2	0.1	0.2	0.4	1.0	0.7	1.1	1.0	0.8	0.4	0.2	0.2		
	DB	-8.1	-0.3	28.3	44.8	73.6	86.1	89.5	82.6	67.2	38.2	26.6	-5.3			
	MCWB	-9.0	-1.2	25.0	35.4	53.0	61.5	66.3	62.9	53.6	34.1	26.0	-6.0			
	2% DB	-14.4	-7.3	21.9	41.1	66.8	81.1	85.5	77.8	59.5	33.7	12.4	-14.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	-14.9	-8.1	18.8	33.1	49.6	59.8	64.2	61.5	49.6	30.6	11.4	-14.6			
	5% DB	-20.9	-14.2	15.3	38.1	61.1	77.1	81.6	73.6	54.8	31.1	2.3	-20.8			
	MCWB	-21.1	-14.8	12.9	31.4	46.5	58.0	63.3	59.9	47.0	28.4	1.7	-21.0			
	10% DB	-27.9	-20.0	9.0	34.5	56.4	73.4	77.8	69.2	50.7	28.4	-5.2	-27.5			
	MCWB	-28.1	-20.3	7.1	29.4	44.4	56.7	61.6	57.6	45.3	26.2	-5.8	-27.8			
	0.4% WB	-8.5	-1.1	25.0	36.5	54.2	64.3	69.0	66.2	54.4	35.7	26.0	-5.9			
	MCDB	-7.9	-0.3	28.6	43.3	71.1	76.8	82.4	77.6	63.4	37.0	26.3	-5.3			
	2% WB	-14.8	-8.0	18.7	33.8	50.4	61.4	66.8	63.0	51.2	31.4	11.5	-14.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	-14.3	-7.3	22.0	39.9	64.1	76.8	80.4	74.1	57.3	33.3	12.3	-14.1			
	5% WB	-21.1	-14.5	12.8	32.0	47.7	59.8	65.0	60.8	48.9	28.4	1.8	-21.0			
	MCDB	-20.8	-14.0	15.2	37.4	59.4	73.9	77.7	70.4	53.4	30.6	2.5	-20.7			
	10% WB	-28.1	-20.3	7.1	29.7	45.2	58.1	63.1	58.7	45.6	26.0	-5.7	-27.7			
	MCDB	-27.9	-20.0	8.9	34.7	55.0	70.9	74.8	67.7	50.0	28.4	-5.1	-27.5			
(35) Mean Daily Temperature Range	MDBR	11.2	18.4	28.7	27.4	21.4	24.5	24.1	24.0	17.0	15.4	13.1	10.0			
	5% DB	19.7	22.2	29.3	24.5	32.1	34.2	34.0	32.8	26.4	14.0	20.0	19.2			
	MCWBR	19.5	21.5	26.1	17.6	17.4	15.5	15.5	17.4	16.3	11.1	19.3	18.8			
	5% WB	19.6	22.2	29.2	21.7	28.3	30.2	28.8	27.8	22.0	13.0	19.9	19.2			
	MCWBR	19.5	21.6	25.9	16.0	15.8	15.1	14.7	15.8	14.5	10.8	19.3	18.7			
(40) Clear-Sky Solar Irradiance	taub	0.179	0.243	0.297	0.364	0.416	0.400	0.493	0.426	0.329	0.261	0.186	0.168			
	taud	1.927	2.083	2.033	1.914	2.018	2.286	1.993	2.165	2.401	2.207	1.919	1.895			
	Ebn at Noon	133	215	245	250	248	259	225	232	240	206	121	59			
	Edh at Noon	9	20	33	47	47	37	48	37	23	17	9	4			
(44) All-Sky Solar Radiation	RadAvg	66	351	941	1583	1697	1722	1603	1250	701	367	117	25			
	RadStd	6	18	45	62	101	148	128	101	84	36	9	2			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page